

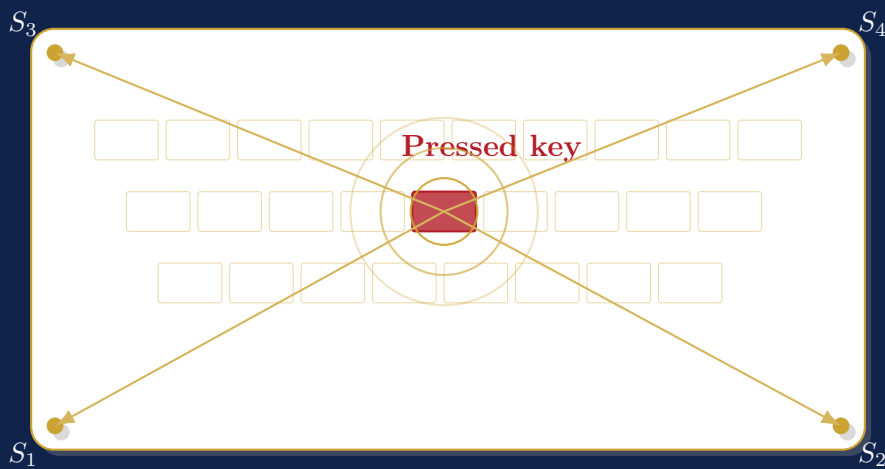
NATIONAL HIGHER SCHOOL OF ADVANCED TECHNOLOGIES

MI2 Second Year Workshop

TRACK 3: THE ATOMIC WHISPER

Nano-Phononics and Acoustic Triangulation

as a Side-Channel Attack Vector



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1. Introduction and Physical Intuition

A localized key press on a keyboard is not only a digital event but also a mechanical event. The force applied by the finger creates a short impulse in the supporting plate, and this impulse propagates through the structure as an elastic disturbance.

If several sensors are placed at known positions, the same event reaches them at slightly different times. These tiny differences contain enough information to estimate the place where the key was pressed.

This report studies how those time differences can be used to reconstruct the position of the source. The approach combines elastic wave propagation, guided waves in thin plates, and geometric localization from time-difference-of-arrival measurements.

Main Concept

When a point on a solid surface is tapped, the generated vibration spreads outward through the material. Points that are closer to the source detect the disturbance earlier than points that are farther away.

Simple Explanation

A key press is not only an action on a keyboard. It also creates a tiny mechanical shake. That shake travels through the plate and reaches the sensors one after another.

Why it Matters

If the wave speed is known, a measured delay can be converted into a distance difference. This is the main idea behind localization.

2. Theoretical Background

2.1. Elastic wave propagation in solids

The propagation of small-amplitude elastic disturbances in a homogeneous isotropic solid is described by the Navier–Lamé equation

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u},$$

where $\mathbf{u}(\mathbf{r}, t)$ is the displacement field, ρ is the density, and λ, μ are the Lamé constants.

This equation expresses how matter deforms and transmits a disturbance after a small mechanical force is applied.

Simple Explanation

This equation is the physics rule that describes how shaking moves inside a solid object. In this project, the shaking starts when a key is pressed.

Example

If one side of a metal tray is tapped, the vibration spreads through the tray. That is a simple example of elastic wave propagation.

2.2. Guided waves in a plate

In a thin plate, the dominant propagating modes are guided Lamb waves. These waves remain confined within the structure and carry the energy emitted by the keystroke.

The anti-symmetric mode A_0 is often important because it is commonly detectable in timing-based localization.

Simple Explanation

A thin plate has special ways of vibrating. Some waves prefer to stay inside the plate and travel along it.

Why it Matters

Knowing the dominant wave mode helps estimate the correct propagation speed and improves the quality of localization.

2.3. Group velocity

The group velocity is defined by

$$v_g = \frac{d\omega}{dk},$$

and represents the speed of the wave packet envelope carrying the detectable signal.

For a thin aluminium plate, a representative value of about 600 m/s is used in the numerical example.

Important Idea

$$d = v_g \Delta t$$

This relation converts a time measurement into a distance or a distance difference.

Simple Explanation

The group velocity is the speed of the useful part of the vibration signal. If the vibration arrives later, it usually means it travelled farther.

Example

If $v_g = 600$ m/s and the signal arrives $10 \mu\text{s}$ later, then

$$d = 600 \times 10 \times 10^{-6} = 0.006 \text{ m} = 6 \text{ mm}.$$

3. Fundamental Relation and Geometric Localization

Let the unknown source position be

$$P = (x, y)$$

and the sensors be located at

$$S_i = (x_i, y_i).$$

The distance from the source to sensor i is

$$d_i = \sqrt{(x - x_i)^2 + (y - y_i)^2}.$$

If the corresponding travel time is t_i , then

$$d_i = v_g t_i.$$

Since the absolute emission time is not generally known, localization is based on time differences of arrival relative to a reference sensor S_1 :

$$\Delta t_{i1} = t_i - t_1.$$

Multiplying by the group velocity gives

$$\delta_{i1} = d_i - d_1 = v_g \Delta t_{i1}.$$

Each equation of the form $d_i - d_1 = \delta_{i1}$ defines a hyperbola. The source is reconstructed as the intersection point of several such hyperbolic constraints.

Simple Explanation

It is not necessary to know the exact instant when the key was pressed. It is enough to compare which sensor detected the vibration first and which one detected it later.

Why it Matters

Using relative times makes the problem easier, because the unknown emission time does not need to be measured directly.

3.1. Signal Correlation for Delay Estimation

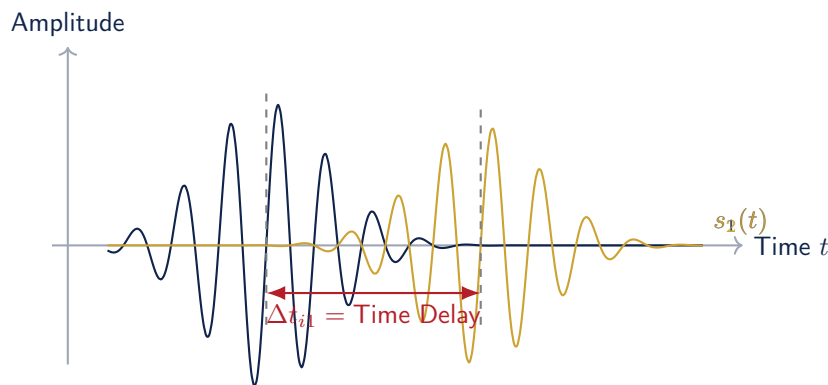


Figure 1: Cross-correlation concept: mathematically sliding the delayed signal $s_2(t)$ against the reference signal $s_1(t)$ to find the precise arrival delay Δt_{i1} .

Simple Explanation

By sliding one signal over the other until their shapes match perfectly (cross-correlation peak), we can precisely find the exact propagation time delay.

4. Model Assumptions

The localization model used in this report relies on several simplifying assumptions:

- the plate is approximately homogeneous and isotropic,
- the dominant signal corresponds to a guided wave packet,
- the effective group velocity v_g is known through calibration,
- sensor coordinates are known accurately,
- first-arrival times can be estimated with sufficient precision,
- boundary reflections and higher-order effects are treated as secondary.

These assumptions simplify the inverse problem while preserving the main physical mechanism.

Simple Explanation

A model is a simplified version of reality. It keeps the main ideas and temporarily ignores smaller effects.

Remark

Real experiments may include reflections, noise, and material imperfections, so calibration and signal processing are important.

5. Geometric and Physical Illustrations

5.1. Surface vibration example

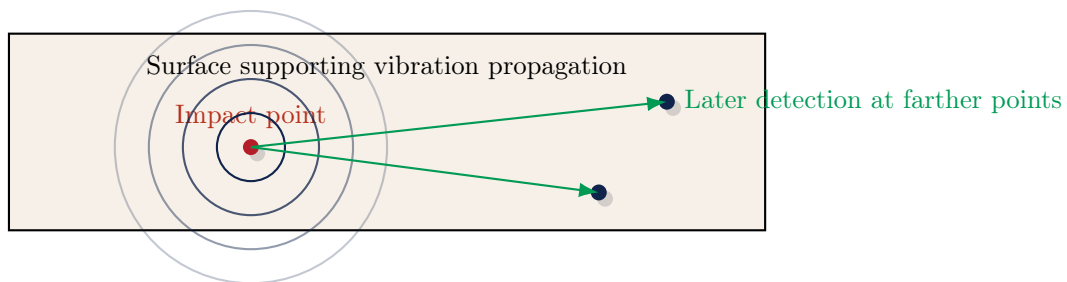


Figure 2: A localized tap on a solid surface generates outward-propagating vibrations.

Simple Explanation

The circles represent the vibration spreading outward with time. Points farther from the impact point detect the signal later.

5.2. Sensor geometry on the plate

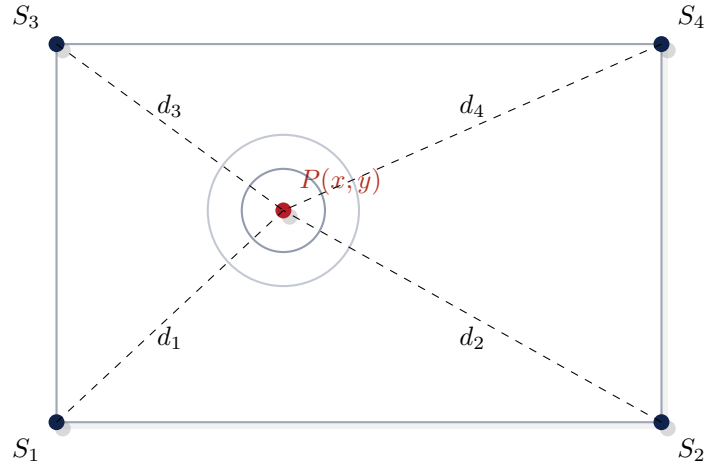


Figure 3: Sensor configuration and unknown source position on a rectangular plate.

Simple Explanation

The red point is the unknown source. The blue points are the sensors used to detect it.

5.3. Hyperbolic interpretation

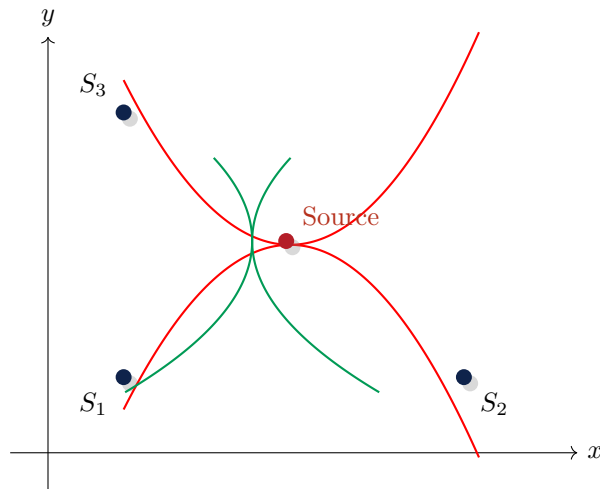


Figure 4: The source position is obtained from the intersection of hyperbolic TDOA constraints.

Simple Explanation

Each delay equation gives one possible curve. The source is located where the different curves intersect.

6. Analytical Solution with Three Sensors

Assume the sensors are placed at

$$S_1 = (0, 0), \quad S_2 = (L, 0), \quad S_3 = (0, L).$$

Define

$$r_1 = \sqrt{x^2 + y^2}, \quad r_2 = \sqrt{(x - L)^2 + y^2}, \quad r_3 = \sqrt{x^2 + (y - L)^2}.$$

The TDOA relations become

$$r_2 - r_1 = \delta_{21}, \quad r_3 - r_1 = \delta_{31}.$$

After squaring and rearranging,

$$x = \frac{L^2 - \delta_{21}^2 - 2\delta_{21}r_1}{2L}, \quad y = \frac{L^2 - \delta_{31}^2 - 2\delta_{31}r_1}{2L}.$$

Substitution into $r_1^2 = x^2 + y^2$ yields a quadratic equation

$$Ar_1^2 + Br_1 + C = 0,$$

whose physical positive root allows the recovery of x and y .

Simple Explanation

This part is the mathematical core of the report. The equations transform time differences into coordinates.

7. Numerical Example

The following parameters are used:

$$L = 0.30 \text{ m}, \quad v_g = 600 \text{ m/s}$$

with sensors

$$S_1 = (0, 0), \quad S_2 = (0.30, 0), \quad S_3 = (0, 0.30)$$

and a true source

$$P = (0.12, 0.18).$$

7.1. True distances

$$r_1 = \sqrt{0.12^2 + 0.18^2} = \sqrt{0.0468} \approx 0.2163 \text{ m}$$

$$r_2 = \sqrt{(0.12 - 0.30)^2 + 0.18^2} = \sqrt{0.0648} \approx 0.2546 \text{ m}$$

$$r_3 = \sqrt{0.12^2 + (0.18 - 0.30)^2} = \sqrt{0.0288} \approx 0.1697 \text{ m}$$

7.2. Distance differences

$$\delta_{21} = r_2 - r_1 = 0.0383 \text{ m}$$

$$\delta_{31} = r_3 - r_1 = -0.0466 \text{ m}$$

7.3. Time differences

$$\Delta t_{21} = \frac{0.0383}{600} \approx 63.8 \mu\text{s}$$

$$\Delta t_{31} = \frac{-0.0466}{600} \approx -77.7 \mu\text{s}$$

7.4. Recovered source

Solving the system gives

$$x \approx 0.120 \text{ m}, \quad y \approx 0.180 \text{ m}.$$

Analysis

In the noiseless case, the reconstructed position matches the true source.

Simple Explanation

A positive delay means a sensor detected the wave later. A negative delay means it detected the wave earlier. These signs already give intuition about the source position.

8. Experimental Pipeline

A complete implementation of the method follows the sequence below:

1. calibration of the plate to estimate the effective group velocity,
2. acquisition of vibration signals from multiple sensors,
3. preprocessing by filtering and normalization,
4. extraction of first-arrival times from the recorded signals,
5. computation of TDOA values,
6. conversion into distance differences using $d = v_g \Delta t$,
7. source reconstruction by analytical or least-squares localization,
8. mapping of the estimated position to the nearest key centroid.

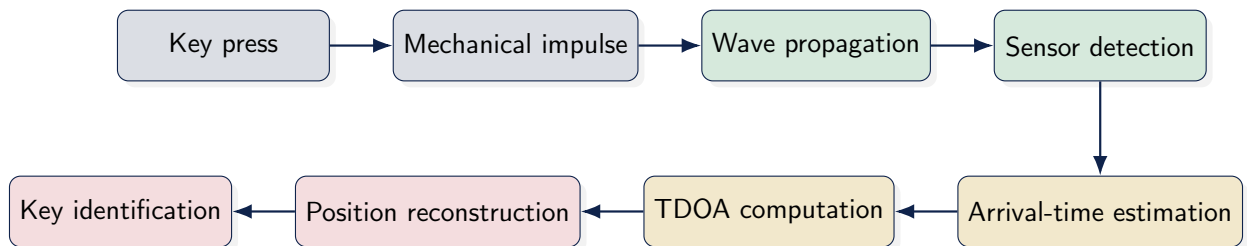


Figure 5: Processing chain from physical interaction to recovered key identity.

Simple Explanation

The system follows a sequence of steps: a key is pressed, the plate vibrates, the sensors detect the signal, time delays are measured, the source position is reconstructed, and the pressed key is identified.

9. Influence of Measurement Noise

The useful approximation

$$\sigma_{xy} \approx v_g \sigma_t,$$

relates the timing uncertainty σ_t to the induced position error σ_{xy} .

For

$$v_g = 600 \text{ m/s}, \quad \sigma_t = 10 \mu\text{s},$$

one obtains

$$\sigma_{xy} = 600 \times 10 \times 10^{-6} = 6 \text{ mm}.$$

Since a standard keyboard key pitch is about 19 mm, this uncertainty is small enough to distinguish adjacent keys with high probability.

Simple Explanation

Noise acts like blur in a measurement. If the blur is much smaller than the distance between two keys, the two keys can still be distinguished.

Remark

If timing uncertainty becomes too large, nearby keys may become difficult to separate correctly.

10. From Position to Digital Key

Once the source position (x, y) has been estimated, it is compared to the coordinates of known key centers. The detected key is the one minimizing the Euclidean distance to the reconstructed position.

This creates a complete chain from the physical domain to the digital domain:

- wave propagation produces the measurable signal,
- geometry provides the location estimate,
- the keyboard map converts the location into a character.

10.1. ASCII, hexadecimal, and binary

For example, the character A can be represented as

$$A \rightarrow 65_{10} \rightarrow 41_{16} \rightarrow 01000001_2.$$

Simple Explanation

The system first estimates a position on the keyboard. Then it selects the nearest key and converts it into digital information.

11. Real-World Case Study: Keyboard Surface Localization

A practical interpretation of the model is a keyboard mounted on a thin support plate. When a user presses a key, the applied force generates a localized mechanical impulse.

That impulse travels through the support structure as a guided elastic disturbance and is detected by sensors attached near the plate boundaries.

Because each sensor is at a different distance from the pressed key, the corresponding arrival times are different. By measuring these delays and applying $d = v_g \Delta t$, the source position can be estimated.

The reconstructed point is then compared with the known keyboard layout in order to infer which key was pressed.

Application

This case study shows how a small physical effect associated with typing can become a measurable information channel.

11.1. Keyboard localization diagram

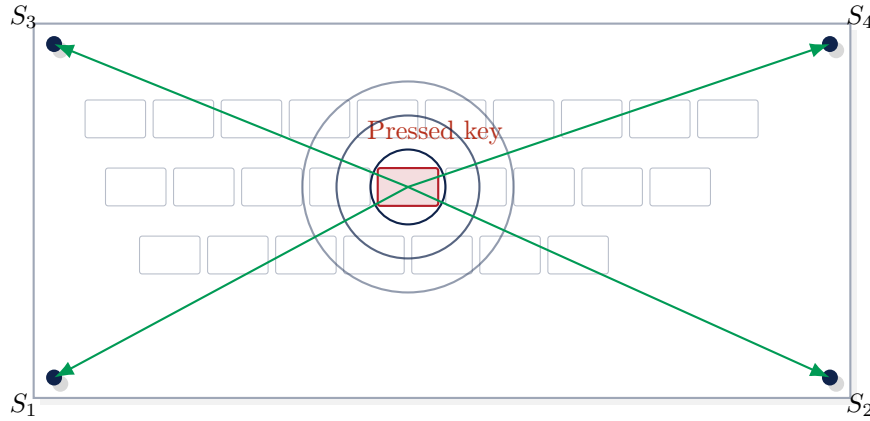


Figure 6: Example of a keyboard mounted on a vibrating plate instrumented with four sensors.

Simple Explanation

The pressed key is shown in red, and the green arrows represent the vibration paths toward the sensors.

11.2. Functional decomposition

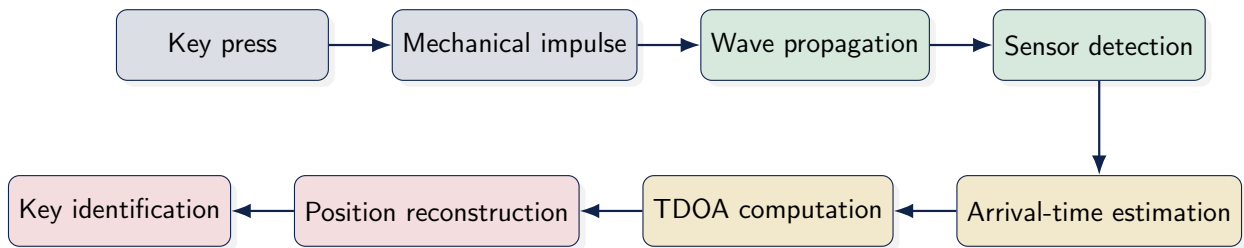


Figure 7: Processing chain from physical interaction to recovered key identity.

Simple Explanation

The whole method follows a clear order: create vibration, detect the signal, measure delays, estimate the position, and identify the key.

12. Additional Real-World Application: Assistive Surface-Tap Interfaces

The same physical principle can also be used in a constructive way. In assistive input systems, a user taps on a surface, sensors capture the resulting vibrations, the localization stage estimates the tap position using $d = v_g \Delta t$, and the detected position is mapped to commands such as select, back, confirm, move left, or move right.

Application

The same localization principle can therefore be interpreted either as a side-channel vulnerability or as a useful accessibility technology.

Simple Explanation

The physics does not change. Only the goal changes: in one case the system reveals private input, and in the other case it helps a user interact with a device.

13. Additional Real-World Example: Smart Door Knock Localization

The same localization method can also be applied to a door panel. When a person knocks on a door, the knock creates vibrations that travel through the wood or composite material.

If sensors are attached at known positions, the system can compare arrival times and estimate where the knock occurred. This can be useful for smart doors, interactive access systems, and knock-based interfaces.

Simple Explanation

Imagine four tiny detectors attached to a door. The one closest to the knock detects the signal first.

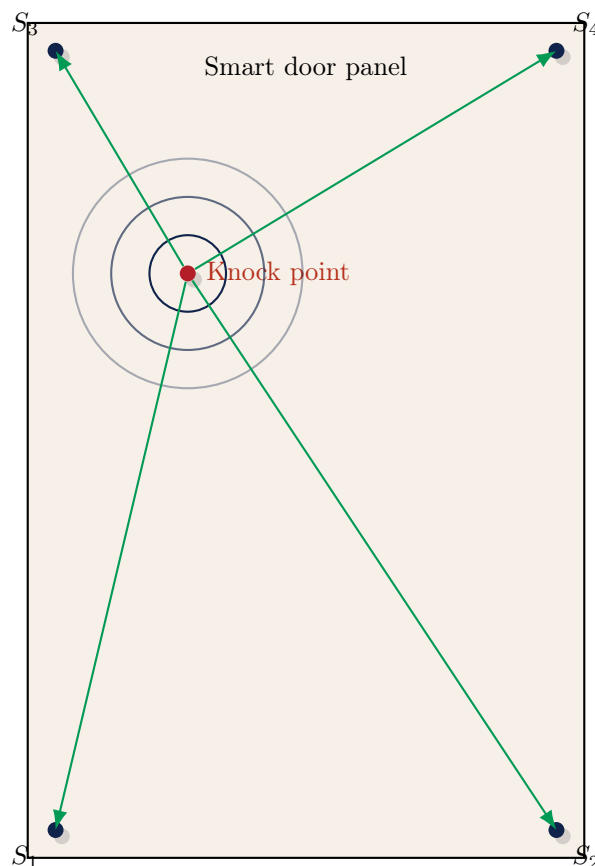


Figure 8: A knock on a door can be localized using the same TDOA principle as a keyboard tap.

14. Third Real-World Example: Smart Wall Tap Localization

Another practical use of the same principle is a smart wall or interactive panel. When a person taps on the wall, the surface vibrates and the vibration spreads through the material.

Sensors fixed at known positions detect the signal at slightly different times. By comparing these arrival times, the system can estimate where the wall was tapped.

This can be used in interactive buildings, museum panels, or touchless control systems where different wall zones correspond to different actions.

Application

This example shows that vibration localization is not limited to keyboards or doors. It can also transform an ordinary wall into an interactive surface.

Simple Explanation

Imagine a wall with invisible buttons. When someone taps one part of the wall, the system listens to the vibration and determines which area was touched.

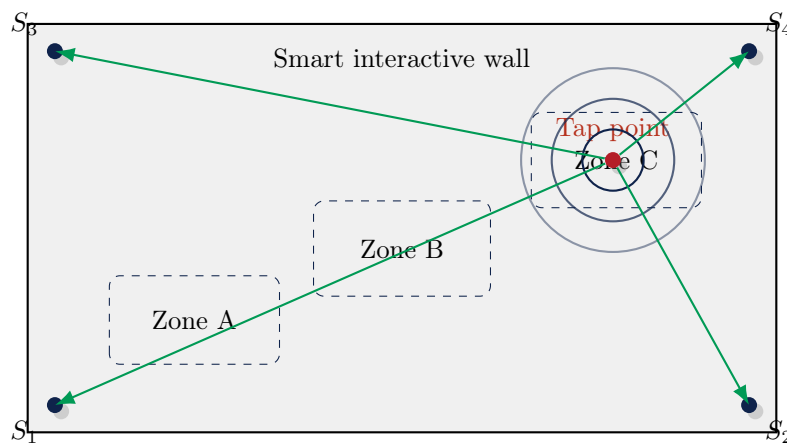


Figure 9: A smart wall can localize a tap and associate it with an interaction zone.

15. Technical Recommendations

Several countermeasures can be considered:

- constrained-layer damping to broaden the wave packet and increase timing uncertainty,
- active acoustic noise injection to reduce the signal-to-noise ratio,
- acoustic isolation joints to attenuate wave transmission,
- software obfuscation to add decoy acoustic events.

Simple Explanation

To make unwanted localization harder, the vibrations can be weakened, blurred, or hidden in noise.

16. Limitations of the Approach

Although the localization method is effective, several practical limitations must be acknowledged:

- guided waves in thin plates are dispersive,
- reflections from boundaries may generate secondary arrivals,
- mechanical coupling between the key and the plate may vary,
- timing uncertainty directly affects localization accuracy,
- material inhomogeneities and mounting conditions may distort propagation paths.

Remark

A real experiment is always less ideal than the mathematical model. This is why calibration and robust signal processing are necessary.

17. Scope and Contribution

This report combines physical modeling, geometric localization, numerical interpretation, and practical engineering analysis. It explains how a localized mechanical disturbance can become a measurable side channel and how the same framework can support constructive applications such as assistive interfaces and interactive surfaces.

The contribution of the study is therefore threefold:

- it clarifies the physical basis of the propagation phenomenon,
- it presents the mathematical structure of source reconstruction,
- it highlights both the vulnerability and the broader application space of vibration-based localization.

Simple Explanation

The report explains the physics, the mathematics, and the practical meaning of the method.

18. Conclusion

This study shows that a keystroke on a solid-supported input surface is not only a digital event but also a measurable mechanical event. The resulting elastic disturbance propagates through the structure with finite group velocity and produces a set of arrival-time differences at spatially separated sensors.

Through the relation $d = v_g \Delta t$, these temporal measurements are converted into geometric constraints that enable source localization.

The analysis demonstrates how wave physics, multilateration geometry, and signal processing can be combined to reconstruct keystroke position with meaningful accuracy. Beyond the side-channel perspective, the same physical principle also supports constructive real-world applications such as vibration-based assistive interfaces, smart doors, and interactive walls.

Important Idea

In one sentence: a small vibration can reveal where an interaction happened.

Simple Explanation

A key press may seem tiny, but it leaves a physical trace. If that trace is measured carefully, its point of origin can be recovered.